

Cyfunke Theorem:

Let $k \in [0, 1]^n$ Define a single hidden layer network as follows:

$$N(x) = \sum_{i=1}^m \alpha_i \sigma(w_i^T x + b_i)$$

$$\alpha_i \in \mathbb{R}, \sigma: \mathbb{R} \rightarrow \mathbb{R}, w_i \in \mathbb{R}^n, x \in \mathbb{R}^n, b_i \in \mathbb{R}$$

for the sigmoid function $\sigma(t) = 1/(1+e^{-t})$ Cyfunke's theorem states that if σ is a continuous and sigmoidal, meaning:

$$\lim_{t \rightarrow \infty} \sigma(t) = 1 \quad \text{and} \quad \lim_{t \rightarrow -\infty} \sigma(t) = 0$$

then the set

$$\mathcal{N}_\sigma = \left\{ \sum_{i=1}^m \alpha_i \sigma(w_i^T x + b_i) : m < \infty \right\}$$

is dense in the space $C[0, 1]^n$

$$\text{that is: } \bar{\mathcal{N}}_\sigma = C[0, 1]^n$$

2 Unif. meas.: $\forall f \in C(k) \forall \varepsilon > 0 \exists N$
s.t. $\sup_{x \in k} |f(x) - N(x)| < \varepsilon$